

Senior Android Engineer Take-Home Assessment

Product Catalog Feature

Overview

Build a Product Catalog feature using Kotlin Multiplatform (KMP) and Clean Architecture. The feature should fetch data from a public API and display it on Android using Jetpack Compose.

Time Expectation: 2 days — focus on quality over quantity.

■■ IMPORTANT: What We're Looking For

We are **NOT** evaluating how beautiful or polished the UI looks. A clean, functional UI is sufficient.

Our primary evaluation criteria are:

- **Architecture** — Proper layer separation, clean boundaries, SOLID principles
- **Code Quality** — Readable, well-organized, idiomatic Kotlin code
- **Testability** — Code designed for easy unit testing, proper dependency injection
- **Maintainability** — Code that's easy to understand, extend, and modify
- **KMP Design** — Shared code structured properly for multiplatform consumption

Requirements

Functional Requirements

Build three screens:

1. Products List Screen

- Display products from the API
- Search products by name
- Pagination support

2. Product Details Screen

- Show full product information when a product is tapped
- Favorite button to save/unsave the product locally

3. Favorites Screen

- Display locally saved favorite products

- Favorites should persist across app launches

Technical Requirements

- Kotlin Multiplatform (KMP) with shared module
- Clean Architecture (proper layer separation)
- Jetpack Compose for Android UI
- Ktor for networking
- Koin for dependency injection
- Local persistence for favorites (candidate's choice)
- Unit tests for business logic (minimum 6 tests)

Note: iOS implementation is not required. However, design your shared ViewModels as if they would be consumed by an iOS app in a real project.

API

Use the free DummyJSON Products API

Base URL: <https://dummyjson.com>

Endpoint	Purpose
GET /products?limit={n}&skip={n}	Paginated list
GET /products/search?q={query}	Search
GET /products/{id}	Product details

Deliverables

1. **Git Repository** — Public repo or grant access to muhammed.peedikayil@Qmobility.ae
2. **README.md** — Setup instructions, architecture overview, and brief iOS considerations (how would iOS consume your shared code?)
3. **Working Android App** — All three screens functional
4. **Unit Tests** — Minimum 6 tests covering mappers, use cases, and ViewModels

Questions?

Contact muhammed.peedikayil@Qmobility.ae

Good luck!